CIE AS PHYSICS (9702) • QUESTION PAPER

7.2 Transverse & Longitudinal Waves

Syllabus Section: 7. Waves **Paper Type:** Structured Questions (Paper 2) **Total:** 5 Questions (44 Marks)

Question 1 Easy [7 marks]

Part (a) [1 mark]

Define a transverse wave.

Part (b) [2 marks]

State **two** properties of a longitudinal wave.

Part (c) [2 marks]

State **two** examples of transverse waves.

Part (d) [2 marks]

State **two** examples of longitudinal waves.

Question 2 Easy [6 marks]

Part (a) [2 marks]

Fig. 1.1 shows graphs of waves X and Y. The scales of the graphs are identical.

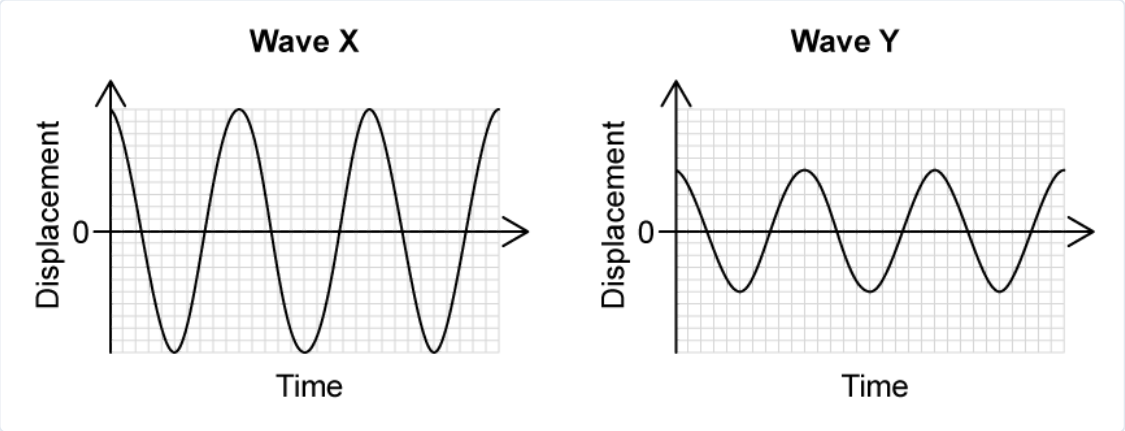


Fig. 1.1

Describe **two** similarities between the properties of waves X and Y.

Part (b) [1 mark]

State **one** difference between waves X and Y.

Part (c) [2 marks]

Distinguish between transverse and longitudinal waves.

Part (d) [1 mark]

Explain how it could be determined whether waves X and Y are transverse or longitudinal.

Question 3 Medium [10 marks]

Part (a) [2 marks]

Explain what is meant by a progressive wave.

Part (b) [4 marks]

An elastic cord is fixed at one end and attached to a mechanical oscillator at the other end. Fig. 1.1 shows, at time t = 0, the shape of a section of the cord as the wave travels from left to right. W, X, Y and Z are four marked points on the cord.

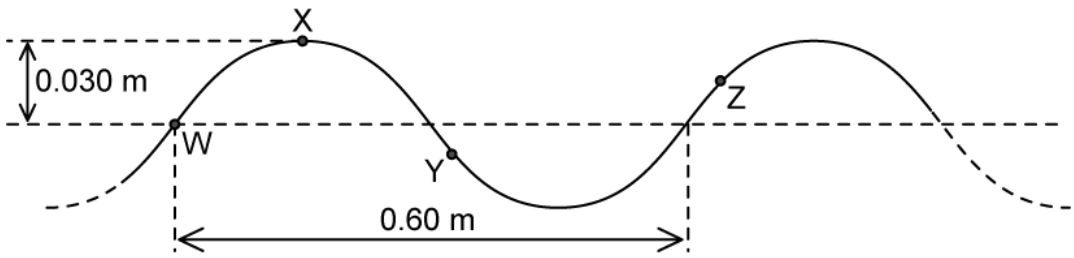


Fig. 1.1

The mechanical oscillator has a steady frequency of 5.0 Hz. The wave has a wavelength of 0.60 m and an amplitude of 0.030 m.

On the axes of Fig. 1.2 sketch the graph of the displacement of point X over the period t = 0 to 0.40 s. Add suitable scales to the axes.



Fig. 1.2

Part (c) [2 marks]

For time t = 0, state which of the points W, X, Y and Z

i) is instantaneously at rest

[1]

ii) has the greatest speed

[1]

Part (d) [2 marks]

The speed of point W on the cord at t = 0 is 0.94 m s^{-1}. With the cord at its original tension, the frequency of oscillation is now doubled to 10 Hz.

The amplitude is kept at 0.030 m. Calculate the new speed of point W at t = 0. Explain your reasoning.

Question 4 Medium [12 marks]

Part (a) [3 marks]

Explain briefly how transmission of energy by a longitudinal wave differs from transmission of energy by a transverse wave. Give **one** example of a transverse wave.

Part (b) [3 marks]

With the aid of a clearly labelled diagram explain how a sound wave in air transmits energy away from its source.

Part (c) [3 marks]

Short pulses of sound are reflected from a wall 30 m from the sound source. The reflected pulses return to the source after 0.18 s.

Calculate the speed of sound.

Part (d) [3 marks]

Fig.1.1 represents the sound wave from part (c).

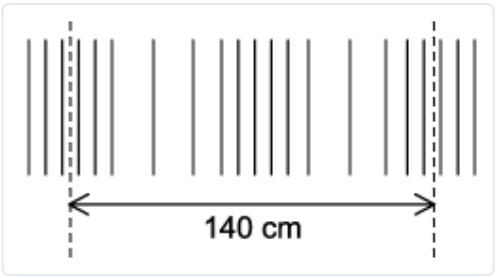


Fig. 1.1

Calculate the frequency of the sound wave.

Question 5 Medium [9 marks]

Part (a) [2 marks]

A large water tank is set up so that a wave can be generated at each end of the tank. The two waves, A and B, travel towards each other at the same speed.

Fig. 1.1 shows the variation of displacement of the water surface with distance travelled at a particular instant.

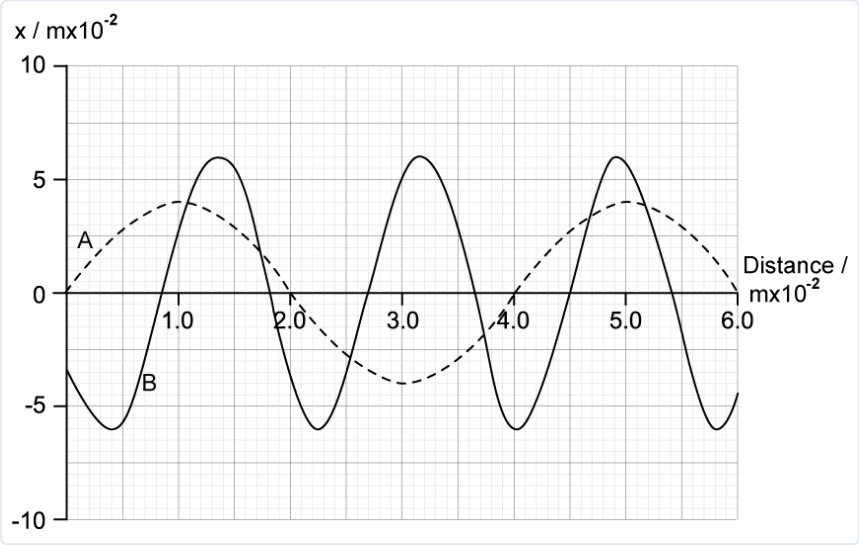


Fig. 1.1

Deduce how many times greater the amplitude of B is to the amplitude of A.

Part (b) [4 marks]

Wave A has a frequency of 8.0 Hz.

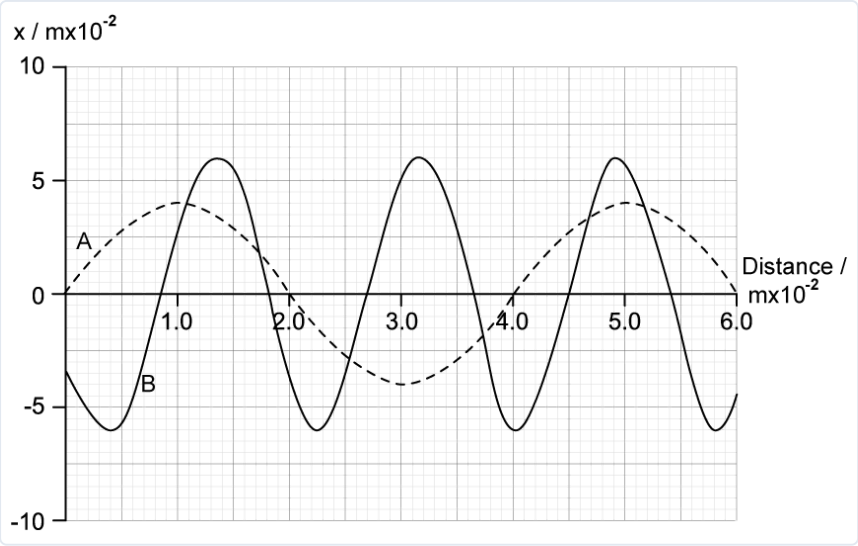


Fig. 1.1

(i) Calculate the velocity of wave A.

[2]

(ii) Determine the frequency of wave B.

[2]

Part (c) **[3 marks]**

Explain how the stationary wave is generated in the tank.